

Indoor environment regulation should consider personalization: Evidence from the evaluation of health effects in the kitchen

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ABSTRACT

Thermal and relative humidity (RH) comfort of the human body is a central indicator for evaluating temperature and RH regulation in indoor environments. However, apart from the focus on thermal comfort, the health effects on the human body should also be taken into consideration. So far, there is insufficient scientific evidence regarding the complete alignment between thermal comfort and health effects in indoor environments. Therefore, we conducted a study to evaluate the impact of water-soluble organic pollutants on human health in kitchen environments under different temperature and RH conditions, by simulating cooking scenarios for residents. We selected four disease probes highly associated with human diseases, namely α -Glucosidase, tyrosinase (TYR), hyaluronidase (HAase), and acetylcholinesterase (AChE), as evaluation indicators. The research results demonstrated significant differences in the inhibitory effects of these four enzymes under different RH levels, with the weakest inhibition observed at high RH. Furthermore, the coupled study on temperature and RH showed that relatively lower temperature and RH had stronger inhibitory effects on TYR, α -Glucosidase, and HAase, while the opposite effect was observed for AChE. Moreover, in the absence of local ventilation, the inhibition rates of these four enzymes were higher. These findings indicate that human thermal comfort and health effects are not always in complete alignment, and for individuals with specific diseases or disease risks, the most suitable thermal and RH comfort environment may not necessarily be the most beneficial for health. Therefore, when regulating temperature and RH in indoor environments, personalized regulation based on health effects should be fully considered.

1. Introduction

Residential indoor environmental quality has an extremely important impact on people's health and quality of life [1,2]. Poor environmental quality can lead to various discomfort symptoms, such as respiratory diseases, headaches, fatigue, and more [3–6]. On the other hand, good indoor environmental quality helps improve people's quality of life and health [7]. Scientists have taken a series of measures to create a good indoor environment. For example, they have developed air purifiers that can filter bacteria and harmful chemicals in the air [8]. At the same time, scientists have studied and improved indoor temperature and relative humidity (RH) control and energy-saving technologies to enable users to maintain a comfortable indoor environment without wasting energy [9,10]. In addition, they have strengthened the monitoring and

assessment of harmful substances and proposed a series of prevention and control measures [11]. For example, researchers have suggested that nanofiber filters could be used to control air pollution in indoor environments [12], a reasonable ventilation layout pattern, rate and other methods can also better control the pollution condition of indoor environment [13]. Some researchers have also suggested that indoor environmental pollution can be prevented by reducing combustion gases from cooking cookers, etc., and using fewer formaldehyde-containing products [14]. Through these efforts, scientists have made contributions to safeguarding people's health and quality of life.

Currently, research on indoor environment mainly focuses on improving indoor air cleanliness, temperature, RH, and enhancing comfort [15,16]. Indoor environment regulation is mostly considered from the perspective of human comfort such as heat and RH, while the

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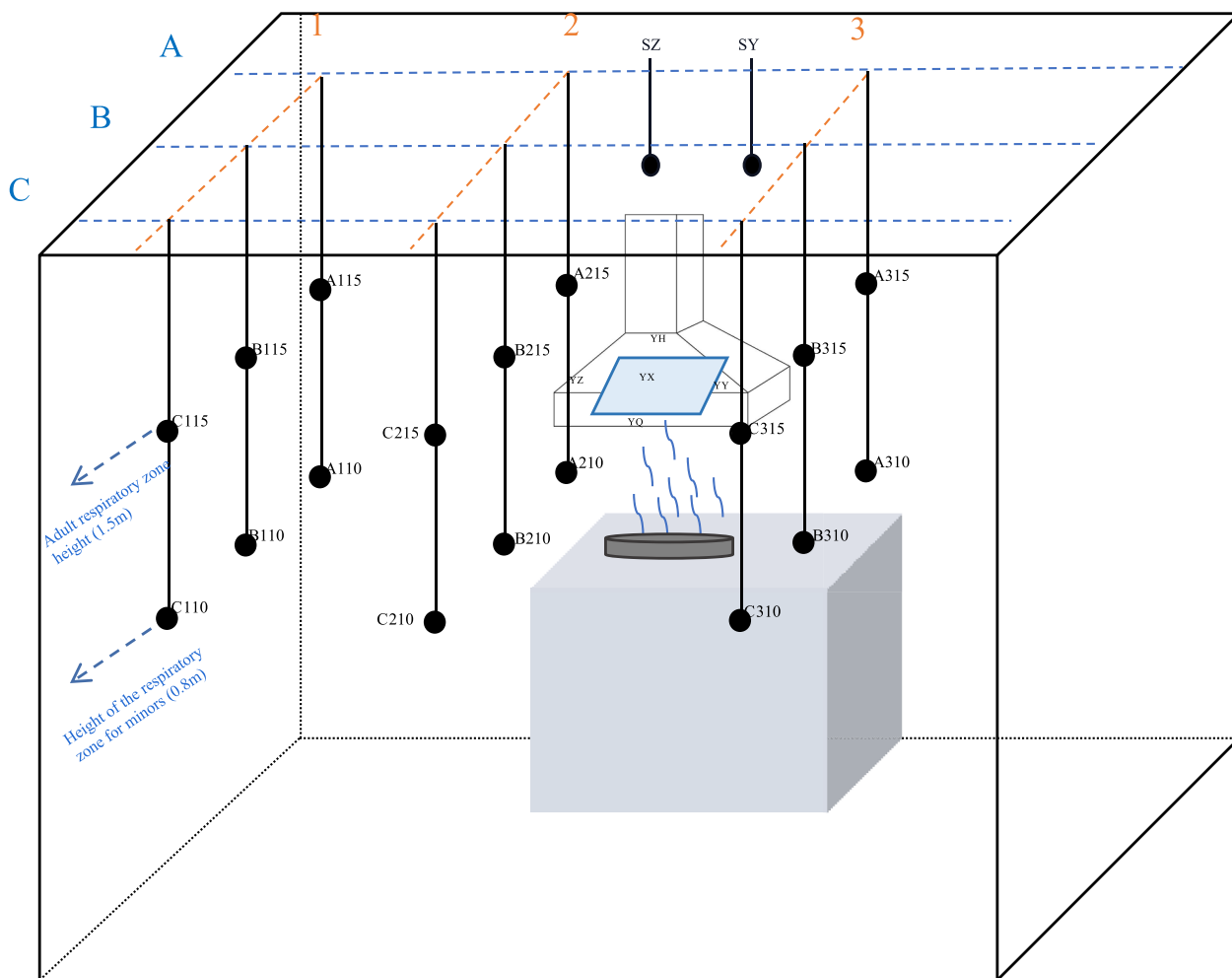


Fig. 1. Schematic diagram of sample point settings. The black points in the figure represent each sampling point, in which YX, YZ, YH, YY and YQ are the sampling points around the local exhaust device. SY, SZ are two sampling points at a height of 1.8 m above the local exhaust device.

real health effects are rarely considered due to the lack of research evidence. In fact, there is a lack of scientific evidence that the body's thermal and wet comfort and health effects are absolutely uniform. In particular, some people with special diseases or disease risk, such as diabetes patients, the elderly, the most suitable heat and RH comfortable indoor environment, must have the most beneficial to health? That's something to be judged by scientific evidence. Therefore, this study takes the kitchen environment as an example to carry out related exploration, in order to provide more scientific suggestions for the regulation of indoor environment.

The kitchen is a relatively special area in the indoor environment. On the one hand, it provides space and carriers for cooking activities [17], on the other hand, it is an important source of pollution in the indoor environment [18]. Volatile organic compounds (VOCs) are one of the most important pollutants in the kitchen environment [19], and most of the traditional research takes total VOCs as the research object. Some studies have shown that RH affects the initial stage of VOCs emissions and is positively correlated with emission concentration [20]. However, we believe that "when the VOCs generated during cooking spread in the indoor air, water-soluble VOCs can dissolve in the water in the air, resulting in a significant change in the retention time of different temperature and RH in the environment, while lipophilic VOCs are relatively less affected. At the same time, after VOCs enter the human respiratory system, the water-soluble components are more easily dissolved, absorbed by the water in the body, participate in metabolism, and affect health." Therefore, paying attention to water-soluble VOCs in

the indoor environment has important theoretical significance and practical value.

In recent years, with the rapid development of modernization, population growth and the accelerated ageing of the population, diabetes now poses a grave concern to the whole world. Worldwide, the number of people with diabetes is rising rapidly [21]. Alzheimer's disease (AD) has also become a rapidly growing global public health problem [22]. In addition, skin health and liver injury and other diseases have gradually become a worldwide concern [23]. Some studies have found that, α -Glucosidase can characterize the human blood sugar level and male reproductive-related diseases [24,25]. Acetylcholinesterase (AChE) are closely related to the transmission of neural signals [26]. Tyrosinase (TYR) is a key metal enzyme in melanin biosynthesis and plays a key role in preventing skin damage caused by ultraviolet radiation [27]. Hyaluronidase (HAase) is a general term for enzymes that can hydrolyze hyaluronic acid, which is widely distributed in the intercellular matrix of various tissues in the human body and has important physiological functions such as maintaining cell osmotic pressure stability and cell adhesion [28], and studies have found that the activity of serum HAase also can provide an early indicator of liver injury [29]. These diseases correspond to a large number of people, and their sensitivity to environmental changes is different from that of normal people, and there is a lack of research on whether the optimal thermal environment and RH environment is the most beneficial to their disease health.

In order to provide experimental scientific evidence for the above viewpoints, this study proposes to focus on the residential kitchen

Table 1
Instruments and reagents.

Instruments or Reagents	Model or Specification	Manufacturer of products
Air dust particle detector	JC-M9	Xuzhou jincheng Environmental Protection Equipment Co., Ltd
Indoor air quality monitor	TSI 7575	TSI Incorporated
Microplate reader	Spectra Max Plus 384	Shanghai Megadragon Molecular Instruments Co., Ltd.
Cultivation Cabinet	BSC-250	Medical equipment factory of Shanghai bosun Industrial Co., Ltd
Analytical balance	AL204	Mettler Toledo Co., Ltd
Locomotive pipette	Research plus	Eppendorf China co., Ltd
Tyrosinase	25 KU,1100 U/mg	Beijing Solarbio Co., LTD.
α -Glucosidase	1 KU,79 U/mg	Beijing Solarbio Co., LTD.
Hyaluronidase	100 mg	Beijing Solarbio Co., LTD.
Acetylcholinesterase	50 mg	Beijing Solarbio Co., LTD.
phosphate buffer solution (PBS)	500 mL	Beijing Solarbio Co., LTD.
4-nitrophenyl- β -D-pyranoglucoside (PNPG)	1 g	Beijing Solarbio Co., LTD.
Sodium Carbonate (Na ₂ CO ₃)	500 g	Beijing Solarbio Co., LTD.
2-nitrobenzoic acid (DNFB)	1 g	Beijing Solarbio Co., LTD.
Acetylthiocholine Iodide (ACTI)	1 g	Beijing Solarbio Co., LTD.
Cetyltrimethylammonium Bromide (CTAB)	25 g	Beijing Solarbio Co., LTD.
Sodium Hyaluronate	1 g	Beijing Solarbio Co., LTD.
Vitamin C	–	Tianjin Bodi Chemical Industry Co., Ltd
Kojic acid	25 g	Shanghai yuanye Bio-Technology Co., Ltd
Sodium hydroxide (NaOH)	500 g	Tianjin Zhiyuan Chemical Co., Ltd
Acetic acid sodium acetate	500 g	Tianjin Zhiyuan Chemical Co., Ltd
Anhydrous ethanol	–	Tianjin Zhiyuan Chemical Co., Ltd
Methanol	–	USA TEDIA

environment, conduct simulation cooking activities in an artificial climate chamber, collect water-soluble VOCs produced, and conduct toxicity effect evaluations with the assistance of disease probes. The disease probes included α -glucosidase, TYR, HAase and AChE, which are highly related to human diseases. At the same time, we also explore the impact of different temperature, RH, and ventilation conditions on toxicity effects. Through this research, we attempt to verify the important value of water-soluble VOCs, and indoor environment control should consider personalized solutions (considering both human comfort and health effects), thereby providing a theoretical basis for personalized creation of healthy indoor environments.

2. Materials and methods

2.1. Experimental platform construction and sampling point layout

A platform is built in the artificial environment laboratory reality to simulate daily kitchen cooking scenarios. The area of the artificial environment room is 12 m² (3.82 m in length, 3.15 m in width) with a height of 2.5 m. It is equipped with a temperature control system (0 ~ 50 °C), a RH control system (20 % ~ 98 %), and a ventilation circulation system. Referring to the Indoor Air Quality (IAQ) Standard GB/T18883-2022, two sampling point heights of 0.8 m and 1.5 m were designed to characterize the minor breathing zone and adult breathing zone, respectively. At the same time, different sampling points were set according to the spatial location and the exhaust device, and their distribution is shown in Fig. 1. The instruments and reagents used in this experiment are shown in Table 1.

Table 2
Temperature, RH, and ventilation parameters of different experiments.

Number of experiments	Exhaust conditions	Temperature (°C)		RH (%)	
		Range of change	Mean	Range of change	Mean
1	–	25.7 ~ 29	27.3	72.3 ~ 82.7	75.7
2	+	24.2 ~ 26.3	25.2	72.4 ~ 81.2	75.1
3	+	18.7 ~ 22.6	20.3	48.7 ~ 67.4	59.1
4	–	18.5 ~ 23.3	20.6	52.9 ~ 67.9	59
5	–	22.9 ~ 25.9	23.9	85.9 ~ 98.3	95.1
6	–	17.7 ~ 21.1	20.0	24.5 ~ 52.6	32.3

Note: “–” means no local exhaust; “+” indicates that local exhaust air exists. Samples collected under the conditions of experiments 4, 5 and 6 were selected to analyze the effect of different RH on enzyme activity. Experiments 1 and 4, 2 and 3 were chosen to represent the effect of different temperature and RH coupling on enzyme activity without and with local exhaust ventilation, respectively. Samples collected from experiments 3 and 4 were chosen to analyze the effect of different exhaust conditions on enzyme activity.

2.2. Research program

2.2.1. Design of cooking pollution sources and experimental influencing factors

The implementation site of this study is in Guiyang, China. Therefore, based on the local dietary characteristics, sour and spicy potato chips were selected as the representative dish for cooking simulation to generate pollution sources. Three consecutive stir-frying were performed for each experimental combination, and the cooking duration was 20 min. To avoid the cross-effects of cooking fuel on the simulated environment, an induction cooker was used as the cooking heat source.

RH, temperature, and ventilation conditions are the key factors that need to be focused on in kitchen environment control. In this study, based on the annual average temperature and RH in Guiyang, a temperature range of 20 ~ 25 °C and a RH range of 30 ~ 100 % were selected for simulation, and the actual temperature and RH were recorded during the experiment. When investigating the effect of RH, under fixed exhaust conditions and temperature range, three RH levels were set: high (85.9 ~ 98.3 %), medium (52.9 ~ 67.9 %), and low (24.5 ~ 52.6 %). When investigating ventilation conditions, the RH (48.7 ~ 67.9 %) and temperature (18.5 ~ 22.6 °C) were fixed, and the difference between the cases with and without local ventilation (emission rate of 10 ~ 15 m³/min) was observed. When investigating the coupling of temperature and RH, the differences between samples obtained from two different temperature and RH ranges were observed under conditions of no local ventilation and with local ventilation, respectively. The specific parameters for each item are shown in Table 2.

2.2.2. Sample collection

Color-changing silica gel was used as the material for collecting water-soluble contaminants, and it was pretreated using chromatographic methanol before the start of the experiment to avoid impurities contained in the silica gel from affecting the accuracy of the experimental results. Before the official start of the experiment, indoor PM_{2.5} and PM₁₀ levels were measured using the JC-M9 airborne dust particle detector. Use an IAQ monitor to measure the TVOC concentration level and temperature and RH to determine whether the simulation platform can carry out experiments. Prepare the ingredients needed for cooking. The composition of each serving is 750 g of potatoes, 75 g of garlic, 150 g of bad chili, 150 g of canola oil, 3 g of salt and 4.5 g of chicken essence. The temperature and RH in the artificial environment laboratory were adjusted to reach the required set temperature and RH range, and the

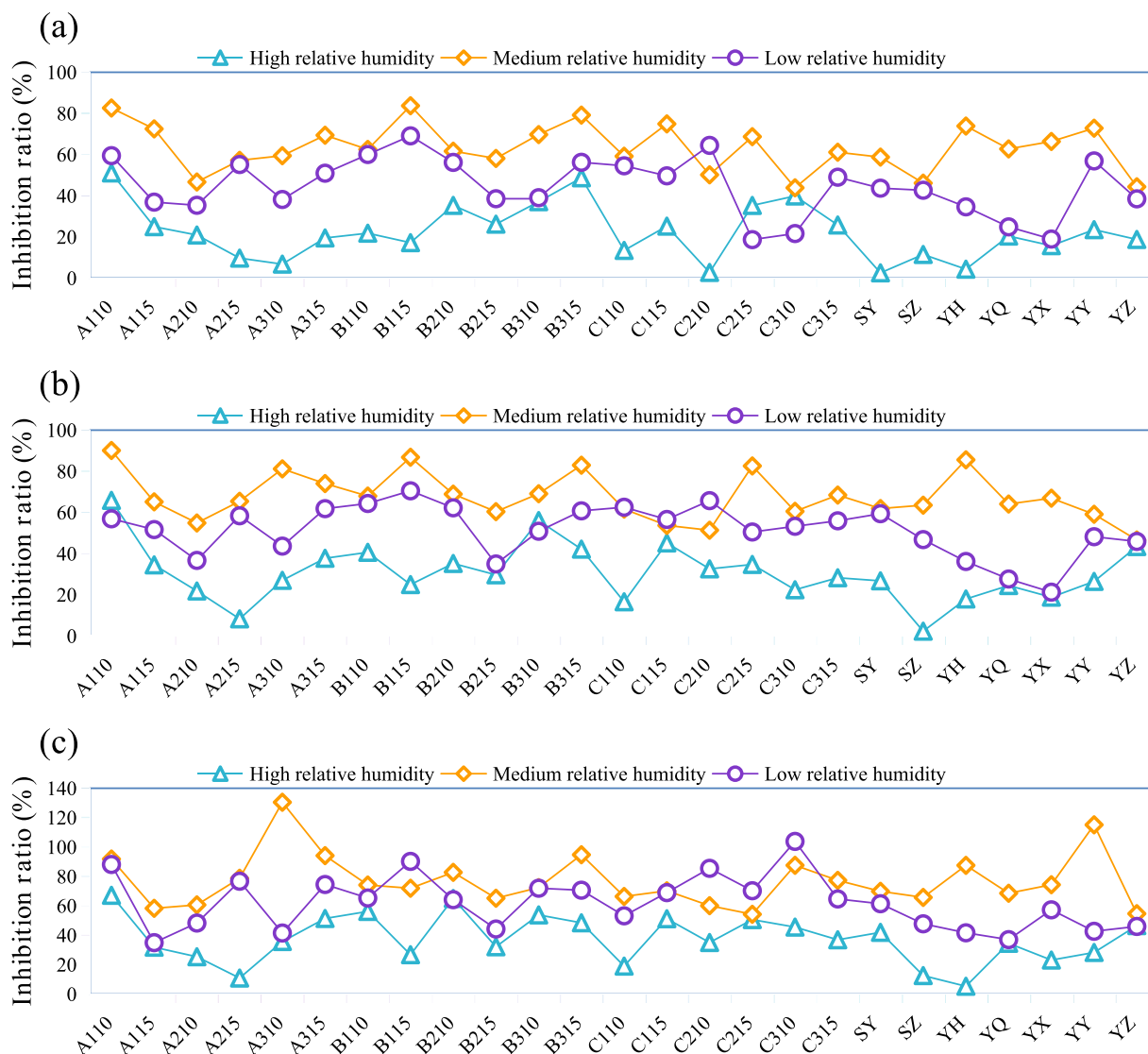


Fig. 2. Inhibition rate of TYR for experimental samples with different RH and concentration. (a) Low concentration of samples, (b) Medium concentration of samples, (c) High concentration of samples.

temperature and RH in the room are recorded. After the room temperature and RH stabilized, 25 pre-treated silica gel, 30 ± 0.2 g each, were quickly weighed and placed in trays on each sampling point, and the relevant test apparatus was turned on to start the experiment. During the experiment, the experimenter stayed in the artificial environment room and started cooking, ensuring the cooking time was about 20 min, and the relevant instruments continuously monitored the temperature and RH of the room during the cooking process. Collect data from the test apparatus after two hours of cooking, collect silica gel samples and seal them, clean the site, open the inner door of the simulation platform for ventilation, and conduct the next experiment after the ventilation is completed.

2.3. Health effect evaluation

Sample Preparation: The sealed silica gel sample of 15 g was accurately weighed by Analytical balance, and placed in a stopper glass test tube. 5 mL of pure water was added into the test tube by Locomotive pipette. After 1 h, the sample was filtered and 2 mL of filtrate was taken into the colorimetric tube for use. The filtrate was mixed with ultrapure water in a certain ratio to make different concentrations (low, medium and high) of the sample to be tested. The specific ratios were 10, 20, and

40 μ L of the renewed filtrate dissolved in 1500 μ L of ultrapure water, respectively. After sample collection, disease probes were constructed for four enzymes, TYR, α -glucosidase, HAase and AchE, respectively, by referring to the method of [30–33], with some minor modifications. The procedure was shown in supplement materials Section 1.

2.4. Statistical analysis

SPSS23.0 software was used for statistical analysis, and the mean $\pm$ variance was calculated based on the original data. Non-parametric tests with 2 or 3 samples of interest were used to analyze the differences between groups. And the difference was statistically significant if $P < 0.05$. SIMCA 13.0 (Umetrics, Umea, Sweden) was used for multidimensional statistical analysis, including unsupervised principal component analysis (PCA), supervised least squares discriminant analysis (PLS-DA), and orthogonal partial least squares discriminant analysis (OPLS-DA). Excel 2021 was used for basic data analysis.

Table 3

The average inhibition rate of four enzymes under different RH conditions.

Enzymes	Concentration	High RH	Medium RH	Low RH
TYR	L	22.34 ± 1.76 %	63.44 ± 1.33 %	44.55 ± 2.01 %
	M	30.58 ± 1.94 %	67.73 ± 1.35 %	51.33 ± 1.53 %
	H	38.04 ± 2.71 %	77.12 ± 3.30 %	77.12 ± 3.30 %
α -glucosidase	L	81.62 ± 0.56 %	97.42 ± 0.30 %	90.39 ± 0.66 %
	M	84.40 ± 0.68 %	104.59 ± 0.32 %	93.54 ± 0.27 %
	H	87.18 ± 0.49 %	110.78 ± 0.77 %	99.97 ± 0.62 %
HAase	L	80.97 ± 0.27 %	102.37 ± 0.42 %	94.02 ± 0.69 %
	M	85.77 ± 0.15 %	105.04 ± 0.36 %	93.11 ± 0.60 %
	H	85.98 ± 0.33 %	108.28 ± 0.81 %	91.06 ± 0.38 %
AChE	L	32.34 ± 1.37 %	47.57 ± 4.07 %	57.84 ± 2.82 %
	M	47.84 ± 3.11 %	61.92 ± 1.76 %	77.89 ± 2.87 %
	H	62.23 ± 2.32 %	84.91 ± 1.34 %	101.68 ± 5.16 %

The 'L', 'M' and 'H' represent low, medium and high concentration samples respectively.

3. Result

3.1. Effect of RH on the activity of different enzymes

The inhibition rate of TYR, α -glucosidase, HAase and AChE by samples collected at each sampling point under the three RH experimental conditions is shown in Fig. 2 and Figs. S1–S3. The average inhibition rate of for enzymes were shown in Table 3. The results showed that the average inhibition of TYR, α -glucosidase, HAase did not increase or decrease linearly with the change of RH under different sample concentration conditions, however, the average inhibition rate of AChE increased with decreasing RH. For TYR, α -glucosidase and HAase, the average inhibition rate of the enzyme was the highest in medium RH, and the lowest in high RH, however, AChE showed the highest average inhibition rate at low RH and the lowest at high RH.

Further analysis was performed using statistical methods and the results showed that, After Bonferroni correction, there was no significant difference in the inhibition of AChE at low sample concentrations between medium and low RH ($P > 0.05$). Apart from this, the other results showed a significant effect of high, medium and low RH samples on the activity of four enzymes ($p < 0.05$) (Tables S1–S4). The results of PLS-DA and OPLS-DA from Fig. 3 showed a clear separation trend for the samples under the three RH conditions of four enzymes. Moreover, there was no significant difference in the effect of samples with different heights of the respiratory zone on the activities of the four enzymes under different RH experimental conditions ($P > 0.05$) (Tables S13–S16).

3.2. Effect of temperature and RH coupling on the activity of different enzymes

In this study, we took TYR, α -glucosidase, HAase and AChE as examples to study the effects of water-soluble organic pollutants in kitchen indoor on the activities of these 4 enzymes under different temperature and RH coupling conditions. The inhibition rate of TYR by samples collected at each sampling point under different temperature and RH coupling conditions is shown in Fig. 4. The inhibition rates of α -glucosidase, HAase and AChE by samples from each sampling point are shown in the Figs. S4–S6. The average inhibition rate of for enzymes were

shown in Table 4. The results showed that, with or without local exhaust, the average inhibition rate of TYR and α -glucosidase was higher when the temperature and RH were relatively lower. However, for HAase, when there is no local exhaust and the temperature and RH are lower, the inhibition rate of HAase is higher, while the opposite is true when there is local exhaust and the temperature and RH are higher. For AChE, with or without local exhaust, the average inhibition rate of AChE was higher when the temperature and RH were relatively higher. In addition, the samples had a concentration effect on TYR, α -glucosidase and AChE, with an increase in enzyme inhibition with increasing sample concentration, whereas HAase did not.

Statistical analysis showed that different temperature and RH coupling conditions had a significant effect on the activity of TYR, α -glucosidase, HAase and AChE ($P < 0.05$) (Tables S5–S8). Figs. S11–S14 showed that the results of PLS-DA and OPLS-DA analysis show that the samples have obvious separation trend under different temperature and RH coupling. Furthermore, when analyzed by respiratory zone height, there was no significant difference ($P > 0.05$) in the effect of samples on TYR, α -glucosidase, HAase and AChE activity under different temperature and RH coupling (Tables S17–S20).

3.3. Effect of ventilation on the activity of different enzymes

The inhibition rate of TYR by samples collected at each sampling point under different ventilation conditions is shown in Fig. 5. The inhibition rates of α -glucosidase, HAase and AChE by samples from each sampling point are shown in the Figs. S7–S9. From Table 5 we can see that, under three kinds of sample concentration, when there is no local ventilation, the average inhibition rate of TYR, α -Glucosidase, HAase and AChE are higher than there is local ventilation.

Statistical analysis showed that the samples with or without local exhaust air had a significant effect on the activity of TYR ($P < 0.05$) (Tables S9–S12). Fig. S10 showed that the results of PLS-DA and OPLS-DA analysis show that the samples have obvious separation trend under different ventilation conditions. In addition, according to the height of breathing zone, samples under different ventilation modes had no significant difference on TYR, α -Glucosidase, HAase and AChE activity ($P > 0.05$) (Tables S21–S24).

4. Discussion

In recent years, people's understanding of residential indoor environment has improved more significantly, and the demand for healthy indoor environment has become more urgent [34]. In this context, the regulation of indoor environment has garnered widespread attention from government officials, scholars, and the general public alike [35–37]. Previously, researchers have prioritized human comfort as a starting point for conducting numerous studies on indoor environmental regulation and exploring the influence of temperature, RH, IAQ, light and other influencing factors on indoor environment quality [38–41]. Additionally, personalized regulation systems have been developed by researchers for these influencing factors, including personalized ventilation and heating systems, etc. [42,43]. It should be noted that these studies have made significant contributions to the regulation of indoor environments. Overall, these studies are essentially conducted from the perspective of the correlation between the built environment and energy. However, we believe that buildings and energy ultimately serve the needs of people, and health is one of the greatest needs of people. When considering buildings and energy, if the health effects are taken into account, it should be a useful experiment or expansion, and even one of the future research trends. Therefore, this study selected kitchen environment (building) as the demonstration scene, constructed four kinds of disease probes (health effects), and evaluated the influence of water-soluble organic pollutants on different enzyme activities from the perspective of temperature and RH regulation (energy).

Our research results showed that in terms of RH, the average

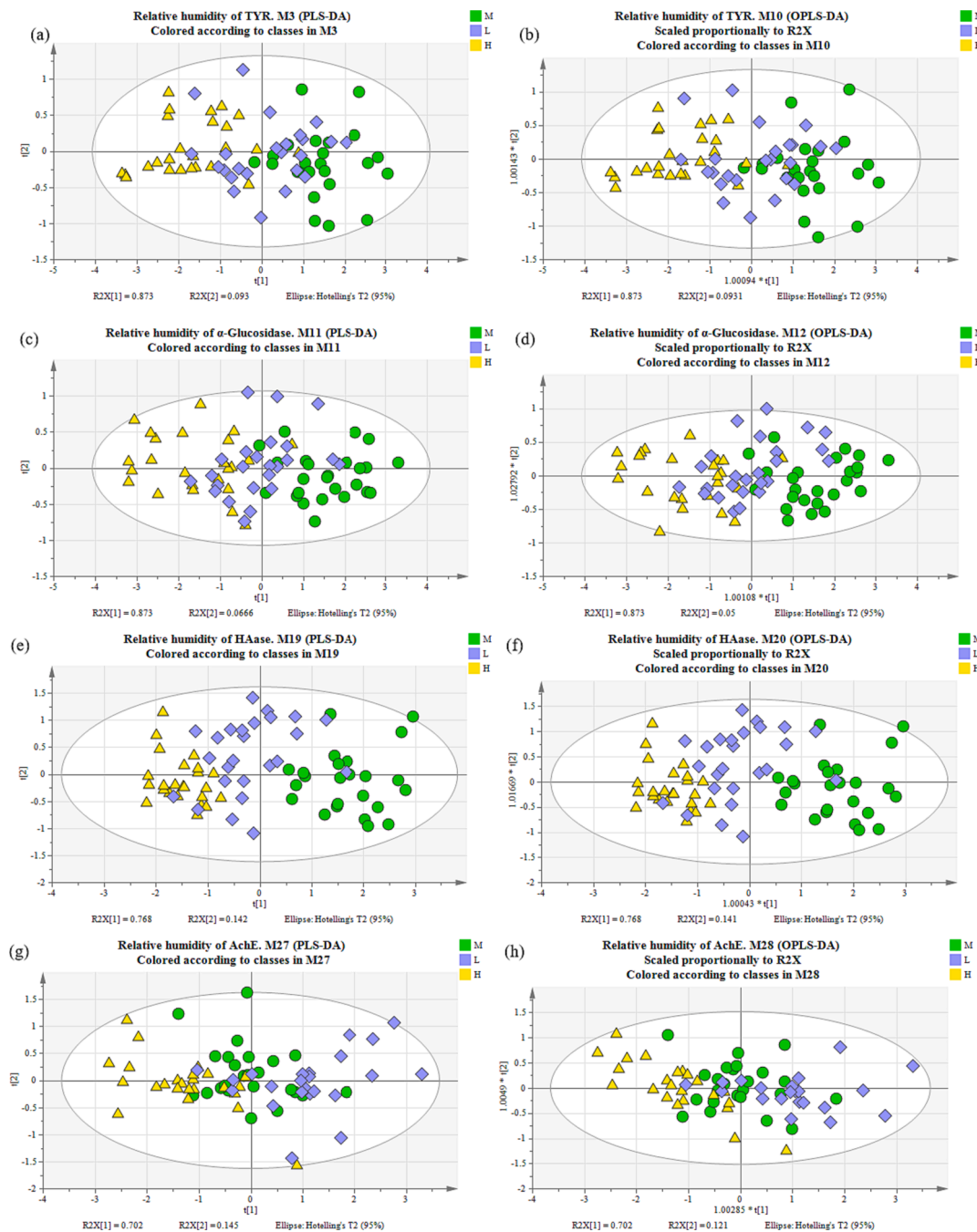


Fig. 3. PLS-DA and OPLS-DA analysis of four enzymes under different RH conditions. (a) PLS-DA of TYR, (b) OPLS-DA of TYR, (c) PLS-DA of α -glucosidase, (d) OPLS-DA of α -glucosidase, (e) PLS-DA of HAase, (f) OPLS-DA of HAase, (g) PLS-DA of AchE, (h) OPLS-DA of AchE.

inhibition rate of TYR, α -glucosidase and HAase was the highest in the medium RH range, while the average inhibition rate of AchE was the highest in the low RH range. In terms of temperature and RH coupling, for TYR and α -glucosidase, regardless of whether there is local exhaust, the inhibition effect of experimental samples with lower temperature and RH is stronger. For AchE, regardless of whether there is local exhaust, the experimental sample with higher temperature and RH has a stronger inhibition effect. These results suggest that the effects of water-soluble organic pollutants on the activities of TYR, α -glucosidase, HAase and AchE in the kitchen indoor environment are not linearly related under different RH and temperature conditions. The reason for this phenomenon may be caused by the inconsistent type and content of water-soluble organic pollution absorbed by tiny droplets of water in indoor air under different temperature and RH conditions. Furthermore,

the results of the research conducted indicate that the suppression of the four enzymes was more significant in the test specimens without local exhaust ventilation as opposed to those with local exhaust ventilation. This can be attributed to the fact that in the presence of local exhaust ventilation, certain water-soluble organic pollutants are expelled outdoors, which ultimately leads to a reduction in the variety and amount of water-soluble organic pollutants absorbed by tiny water droplets in the indoor atmosphere.

The treatment of diabetes is the greatest challenge facing humanity in the 21st century. By 2021, the number of patients has reached 536.6 million [44]. Numerous studies have shown that α -glucosidase activity is associated with blood glucose levels and male reproductive diseases [45,46]. In the small intestine, α -glucosidase is responsible for hydrolyzing carbohydrates into monosaccharides, which enter the

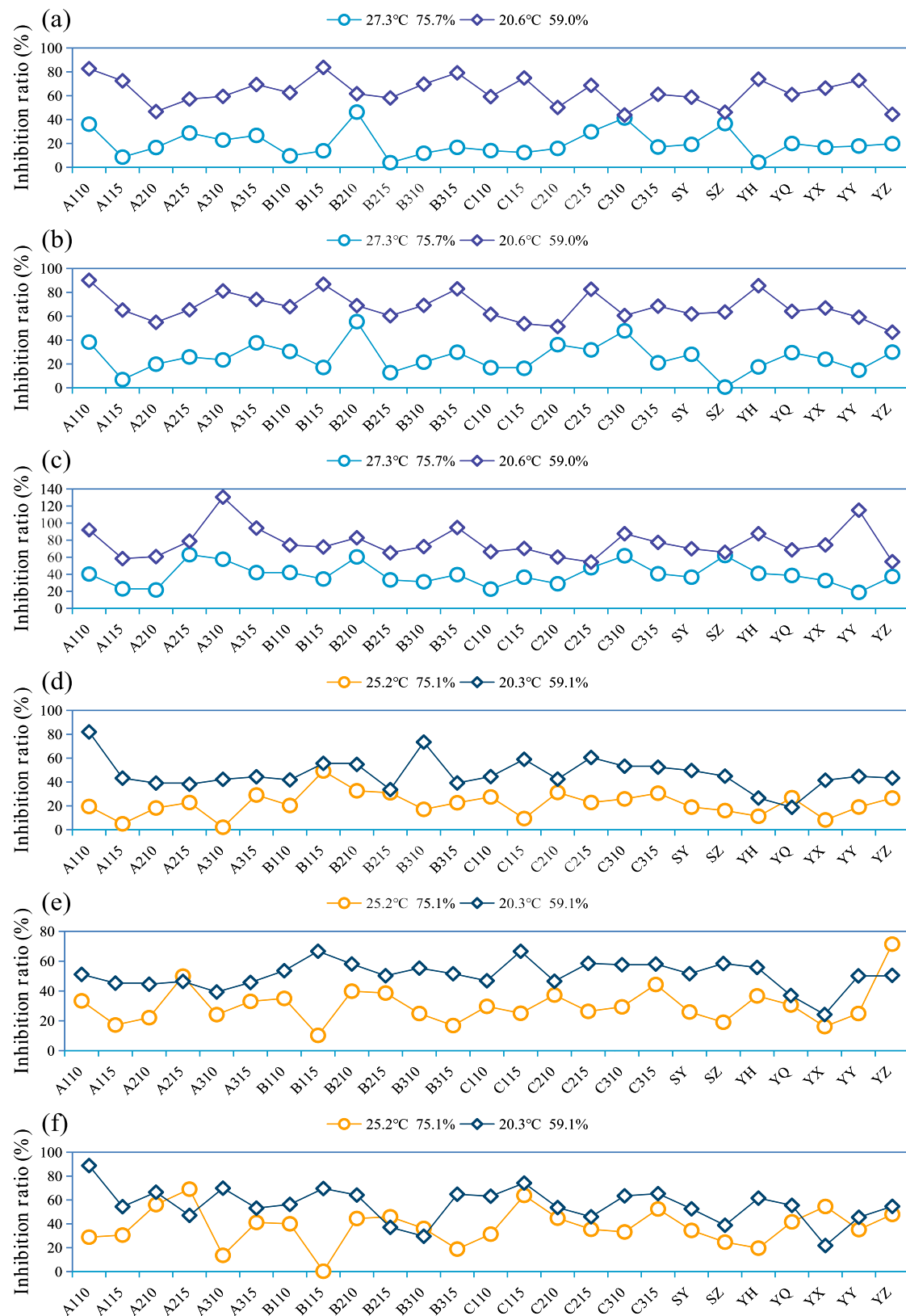


Fig. 4. Inhibition rate of TYR under different temperature and RH coupling conditions and concentration experimental samples. When the temperature and RH were 27.3 °C and 75.7%, and 20.6 °C and 59.0%, respectively, the local exhaust system was not turned on. When the temperature and RH were 25.2 °C and 75.1%, and 20.3 °C and 59.1%, respectively, the local exhaust system was turned on. Low concentration samples: (a) and (d). Medium concentration samples: (b) and (e). High concentration sample: (c) and (f).

Table 4

The average inhibition rate of four enzymes under different temperature and RH coupling conditions.

Enzymes	Concentration	–		+	
		27.3°C	20.6°C	25.2°C	20.3°C
		75.7 %	59.0 %	75.1 %	59.1 %
TYR	L	20.28 ± 1.17 %	63.37 ± 1.28 %	21.77 ± 0.99 %	46.77 ± 1.68 %
		25.34 ± 1.42 %	67.73 ± 1.29 %	30.49 ± 1.55 %	50.79 ± 0.80 %
	M	39.72 ± 1.61 %	77.12 ± 3.17 %	37.77 ± 2.34 %	55.90 ± 2.04 %
α -glucosidase	L	74.96 ± 0.22 %	97.42 ± 0.28 %	84.37 ± 0.33 %	92.24 ± 0.55 %
		79.65 ± 0.37 %	104.59 ± 0.30 %	86.86 ± 0.58 %	95.14 ± 0.38 %
	M	81.72 ± 0.67 %	110.78 ± 0.73 %	93.00 ± 0.73 %	101.37 ± 1.04 %
HAase	L	99.42 ± 0.86 %	102.37 ± 0.41 %	99.56 ± 0.60 %	98.17 ± 0.48 %
		96.03 ± 1.07 %	105.04 ± 0.35 %	102.92 ± 0.53 %	97.21 ± 0.68 %
	M	95.71 ± 0.55 %	108.28 ± 0.78 %	101.21 ± 0.66 %	93.56 ± 0.86 %
AChE	L	64.98 ± 3.36 %	47.57 ± 3.90 %	59.94 ± 5.25 %	38.14 ± 2.65 %
		81.06 ± 4.97 %	61.92 ± 1.69 %	74.87 ± 2.77 %	44.72 ± 2.33 %
	M	97.74 ± 5.04 %	84.91 ± 1.29 %	86.77 ± 4.62 %	59.87 ± 4.11 %

The 'L', 'M' and 'H' represent low, medium and high concentration samples respectively.

bloodstream and cause hyperglycemia [47]. Therefore, inhibiting the activity of α -glucosidase can lower blood sugar levels [48]. AD is a chronic, irreversible neurodegenerative disorder of the central nervous system that causes progressive memory loss [49]. The World Alzheimer's Disease Report estimates that there are 50 million instances of AD worldwide, and that number is anticipated to rise exponentially to 152 million cases by 2050 [50]. AChE is associated with neurodegenerative diseases such as AD [51]. Currently, the prevailing hypothesis regarding the pathogenesis of AD suggests that excessive activity of AChE leads to a decrease in acetylcholine concentration, which in turn results in the degeneration of the cholinergic system in AD patients [52]. HAase is an enzyme that degrades hyaluronic acid, which can indirectly affect physiological functions such as osmotic pressure stabilization of cells. Researchers have discovered six types of HAase in various organs (liver, spleen, eyes, skin, testes, kidneys, and uterus) and body fluids (tears, blood, and semen) [53]. HAase plays a role in wound healing, inflammation, immune response, and fertilization process [54–56]. Changes in the activity of HAase in the human body may trigger this series of diseases. TYR is a key enzyme in melanin synthesis in living organisms and can indirectly affect pigmentation diseases in the body [57]. However, excessive production of melanin might cause considerable problems of esthetic concern, such as hyperpigmentation dermatosis and melanoma [58].

Changes in the activities of these enzymes may cause effects on human health. Different populations respond differently to changes in TYR, α -glucosidase, HAase or AChE activity. For example, activation of α -glucosidase activity can be catastrophic for diabetics or those at high risk for diabetes, leading to exacerbations or significantly increased risk of diabetes in diabetics. For people at risk for hypoglycemia, moderate activation of α -glucosidase activity may be beneficial, whereas inhibition of α -glucosidase activity may significantly increase the risk of hypoglycemia [59]. As another example, for AChE activity, if its activity is significantly inhibited, it will lead to a significant increase in the risk of many neurological diseases [60]; if its activity is significantly activated, it will lead to a significant deterioration in the condition of elderly patients with dementia, and the risk of disease in elderly people without

Alzheimer's disease will be significantly increased [61]. It can be seen that the level of disease-related enzymes required by different groups of people is different, and the health effect needs of different groups of people should be fully considered when regulating the indoor environment. Except the evidence provided in this study, [62] showed that high RH indoors may lead to a significant decline in lung function in the elderly population. Wolkoff's study found that increased indoor RH appeared to reduce nasal symptoms in patients with obstructive apnea [63]. And another research found that an increase in temperature and RH significantly elevated the incidence of hand, foot, and mouth disease [64]. Furthermore, the installation of exhaust fans in the kitchen can reduce the risk of cardiovascular disease [65]. Good kitchen ventilation can reduce the prevalence of depression and anxiety disorders [66].

From the perspective of human comfort, many studies suggest that RH should be between 30 % and 70 % [67–69]. Some studies have also shown that higher RH leads to lower sleep efficiency [70], under low RH conditions, the moisture of the surface skin of the human body drops rapidly and it feels more fatigued at less than 70 % [71]. And a mechanical ventilation in the kitchen environment may be an effective way to improve sleep quality [72]. As a result, there is no complete unity between human health effects and comfort, therefore, it is very necessary to carry out the design of individualized control programs for different people and different disease risks when creating indoor environments. In addition, this study also set sampling points at two heights of 0.8 m and 1.5 m according to the height of the breathing zone, in order to simulate whether there are differences in the response of adults and children to different enzymes. The results of this study showed that samples from different respiratory zone height sampling points had no significant effect on enzyme activity ($P > 0.05$), thus suggesting that water-soluble contaminants in the kitchen environment pose the same health risk to adults and children, and that access by children should be minimized during kitchen use.

Of course, there are some limitations of this study. Due to the huge workload involved, the internal mechanism of TYR, α -glucosidase, HAase or AChE activity changes caused by different RH, temperature, ventilation, etc., is still unclear, that is, the type or content of which toxic substances changes have not yet been clarified. At the same time, the specific indoor environment regulation scheme based on these four enzymes as probes is still incomplete. In the future, we will carry out more in-depth research around these scientific issues, and we expect more scholars to join us to work together on personalized regulation of indoor environment based on health effects, so as to provide a healthier indoor environment for human beings.

5. Conclusion

This study developed a disease probe based on the bio-enzyme activity and explored the intrinsic correlation between water-soluble organic pollutants in the kitchen environment and human health effects from the perspective of human health effects. The results showed that different influencing factors and their levels had different impacts on enzyme activity. This study provides scientific evidence for personalized control schemes to be considered from a health perspective in addition to temperature and RH comfort when regulating indoor environments.

The effect of TYR: in terms of RH, the sample has the greatest average inhibition of TYR at medium RH conditions and lowest at high RH conditions. With or without local exhaust, the average inhibition rate of TYR is always higher at low temperature and RH, and the higher the sample concentration, the stronger the inhibition of TYR. From the perspective of exhaust conditions, the samples collected without local exhaust always have a higher inhibition rate on TYR.

The effect of α -Glucosidase: in terms of RH, the sample has the greatest average inhibition of α -Glucosidase at medium RH conditions and lowest at high RH conditions. With or without local exhaust, the average inhibition rate of α -Glucosidase is always higher at low

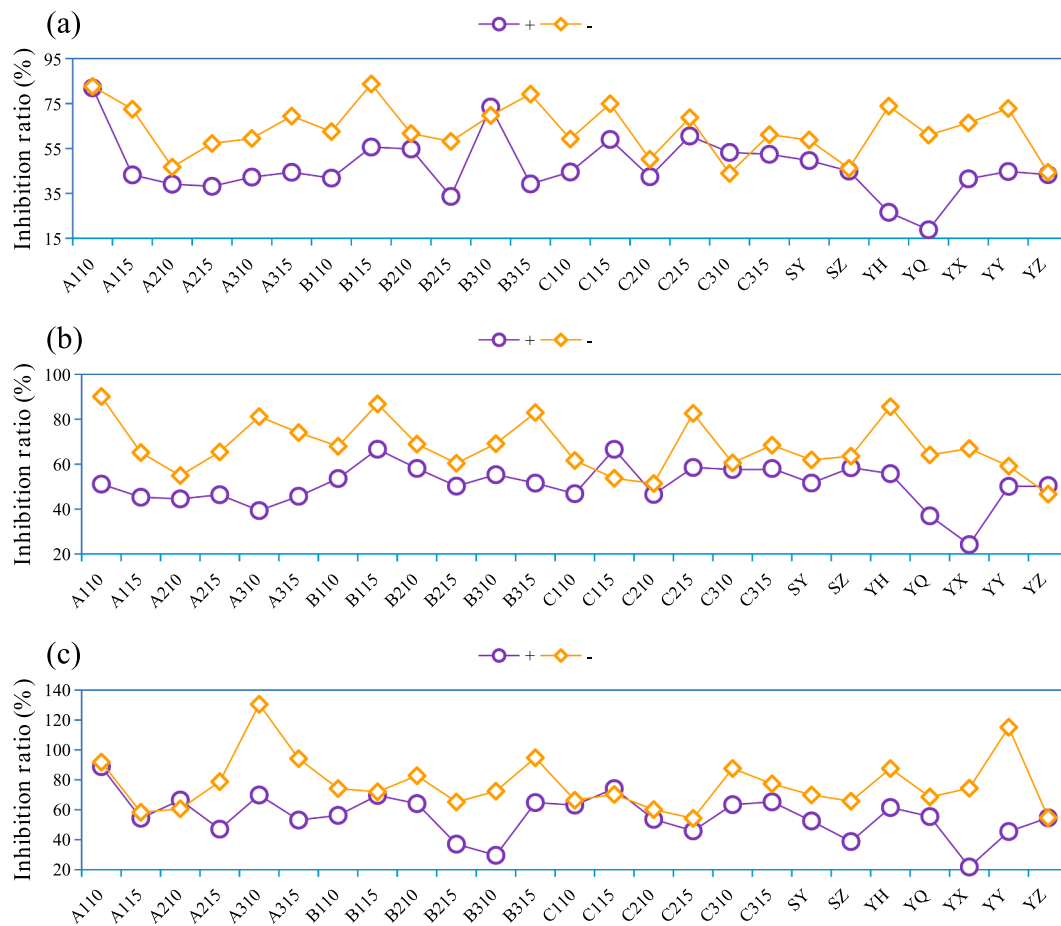


Fig. 5. Inhibition rate of TYR for experimental samples with different local exhaust conditions and concentration. (a) Low concentration sample, (b) Medium concentration sample, (c) High concentration sample.

Table 5

The average inhibition rate of four enzymes under different ventilation conditions.

Enzymes	Concentration	–	+
TYR	L	63.44 ± 1.33 %	46.77 ± 1.75 %
	M	67.73 ± 1.35 %	50.79 ± 0.84 %
	H	77.12 ± 3.30 %	55.90 ± 2.13 %
α-glucosidase	L	97.42 ± 0.30 %	90.24 ± 0.58 %
	M	104.59 ± 0.32 %	95.14 ± 0.40 %
	H	110.78 ± 0.77 %	101.37 ± 1.08 %
HAase	L	102.37 ± 0.42 %	98.17 ± 0.55 %
	M	105.04 ± 0.36 %	97.21 ± 0.71 %
	H	108.28 ± 0.81 %	93.56 ± 0.90 %
AchE	L	47.57 ± 4.07 %	38.14 ± 2.76 %
	M	61.92 ± 1.76 %	44.72 ± 2.42 %
	H	84.91 ± 1.34 %	59.87 ± 4.28 %

The 'L', 'M' and 'H' represent low, medium and high concentration samples respectively.

temperature and RH, and the higher the sample concentration, the stronger the inhibition of α-Glucosidase. From the perspective of exhaust conditions, the samples collected without local exhaust always have a higher inhibition rate on α-Glucosidase.

The effect of HAase: in terms of RH, the sample has the greatest average inhibition of HAase at medium RH conditions and lowest at high RH conditions. When there is no local exhaust, the sample has a stronger inhibitory effect on HAase when the temperature and RH are relatively lower, while the opposite is true when there is local exhaust. From the perspective of exhaust conditions, the samples collected without local

exhaust always have a higher inhibition rate on HAase.

The effect of AchE: unlike TYR, α-Glucosidase, and HAase, in terms of RH, the sample has the greatest average inhibition of AchE at low RH conditions and lowest at high RH conditions. With or without local exhaust, the average inhibition rate of AchE is always lower at low temperature and RH, and the higher the sample concentration, the stronger the inhibition of AchE. From the perspective of exhaust conditions, the samples collected without local exhaust always have a higher inhibition rate on AchE.

CRediT authorship contribution statement

Hongwei Lou: Writing – original draft, Visualization, Software, Methodology, Data curation. **Qinghong Liu:** Writing – original draft, Visualization, Software, Methodology, Data curation. **Huaguo Chen:** Conceptualization, Formal analysis, Investigation, Project administration, Validation, Writing – review & editing. **Qin Yang:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Project administration, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enbuild.2023.113641>.

References

- [1] L. Deng, Q. Deng, The basic roles of indoor plants in human health and comfort, *Environ. Sci. Pollut. Res. Int.* 25 (36) (2018) 36087–36101.
- [2] C. Shrubsole, S. Dimitroulopoulou, K. Foxall, B. Gadeberg, A. Doutsis, IAQ guidelines for selected volatile organic compounds (VOCs) in the UK, *Build. Environ.* 165 (2019) 106382–106405.
- [3] Y. Ni, G. Shi, J. Qu, Indoor PM_{2.5}, tobacco smoking and chronic lung diseases: A narrative review, *Environ. Res.* 181 (2020) 108910–108919.
- [4] P. Fu, Z. Zhao, D. Norback, X. Zhang, K.K.L. Yung, Associations between indoor environment and lifestyles and sick building syndrome symptoms among adults in Taiyuan and Urumqi of China, *Indoor Air* 32 (7) (2022) e13081–e13095.
- [5] N. Suzuki, H. Nakaoka, Y. Nakayama, K. Tsumura, K. Takaguchi, K. Takaya, A. Eguchi, M. Hanazato, E. Todaka, C. Mori, Association between sum of volatile organic compounds and occurrence of building-related symptoms in humans: A study in real full-scale laboratory houses, *Sci. Total Environ.* 750 (2021) 141635–141641.
- [6] Y. Fang, C. Liu, C. Zhao, H. Zhang, W. Wang, N. Zou, A Study of the Effects of Different Indoor Lighting Environments on Computer Work Fatigue, *Int. J. Environ. Res. Public Health* 19 (11) (2022) 6866–6883.
- [7] B.W. Dionova, M.N. Mohammed, S. Al-Zubaidi, E. Yusuf, Environment indoor air quality assessment using fuzzy inference system, *ICT Express* 6 (3) (2020) 185–194.
- [8] B. Zhou, T. Liu, S. Yi, Y. Huang, Y. Guo, S. Huang, C. Zhou, R. Zhou, H. Cao, Reducing the Effectiveness of Ward Particulate Matter, Bacteria and Influenza Virus by Combining Two Complementary Air Purifiers, *Int. J. Environ. Res. Public Health* 19 (16) (2022) 10446–10457.
- [9] H. Zhang, H. Yoshino, Analysis of indoor humidity environment in Chinese residential buildings, *Build. Environ.* 45 (10) (2010) 2132–2140.
- [10] B. Dong, V. Prakash, F. Feng, Z. O'Neill, A review of smart building sensing system for better indoor environment control, *Energ. Buildings* 199 (2019) 29–46.
- [11] T. Salthammer, Emerging indoor pollutants, *Int. J. Hyg. Environ. Health* 224 (2020) 113423–113434.
- [12] D.S. de Almeida, L.D. Martins, M.L. Aguiar, Air pollution control for indoor environments using nanofiber filters: A brief review and post-pandemic perspectives, *Chem. Eng. J. Adv.* 11 (2022) 100330–100337.
- [13] F. Davardoost, D. Kahforoushan, Evaluation and investigation of the effects of ventilation layout, rate, and room temperature on pollution dispersion across a laboratory indoor environment, *Environ. Sci. Pollut. Res.* 26 (6) (2019) 5410–5421.
- [14] J. González-Martín, N.J.R. Kraakman, C. Pérez, R. Lebrero, R. Muñoz, A state-of-the-art review on indoor air pollution and strategies for indoor air pollution control, *Chemosphere* 262 (2021) 128376–128391.
- [15] P. Šujanová, M. Rychtáriková, T. Sotto Mayor, A. Hyder, A Healthy, Energy-Efficient and Comfortable Indoor Environment, a Review, *Energies* 12 (8) (2019) 1414–1450.
- [16] F.-Y. Zhao, J. Cheng, B. Liu, Z.-R. Huang, X. Li, Indoor airflow and pollutant spread inside the cleanroom with micro-porous supplying panel and different ventilation schemes: Analytical, numerical and experimental investigations, *J. Build. Eng.* 31 (2020) 101405–101417.
- [17] S. Liu, J. Dong, Q. Cao, X. Zhou, J. Li, X. Lin, K. Qing, W. Zhang, Q. Chen, Indoor thermal environment and air quality in Chinese-style residential kitchens, *Indoor Air* 30 (2) (2020) 198–212.
- [18] C.A. Alves, E.D. Vicente, M. Evtugina, A.M.P. Vicente, T.A. Sainnkhon, N. Kovats, Cooking activities in a domestic kitchen: Chemical and toxicological profiling of emissions, *Sci. Total Environ.* 772 (2021) 145412–145425.
- [19] H. Wang, Z. Xiang, L. Wang, S. Jing, S. Lou, S. Tao, J. Liu, M. Yu, L. Li, L. Lin, Y. Chen, A. Wiedensohler, C. Chen, Emissions of volatile organic compounds (VOCs) from cooking and their speciation: A case study for Shanghai with implications for China, *Sci. Total Environ.* 621 (2018) 1300–1309.
- [20] H. Shao, Y. Ren, Y. Zhang, C. Wu, W. Li, J. Liu, Factor analysis of the influence of environmental conditions on VOC emissions from medium density fibreboard and the correlation of the factors with fitting parameters, *RSC Adv.* 11 (42) (2021) 26151–26159.
- [21] T. Hasan, A. Islam, R.a.k. Riva, M.N. Rahman, S. Ahmed, M.A. Islam, A.F.M.S. U. Daula, Phytochemicals from Zingiber capitatum rhizome as potential α -glucosidase, α -amylase, and glycogen phosphorylase inhibitors for the management of Type-II diabetes mellitus: Inferences from in vitro, in vivo and in-silico investigations, *Arab. J. Chem.* 16 (10) (2023) 105128–105137.
- [22] R. Logesh, S.V. Sathasivampillai, A triterpenoid saponin bacoside-A3 from the aerial parts of *Bacopa monnieri* (L.) Wettst. with acetylcholinesterase enzyme combating Alzheimer's disease, *S. Afr. J. Bot.* 156 (2023) 177–185.
- [23] H. Zhou, T. Li, B. Li, S. Sun, Skin health properties of *Paeonia lactiflora* flower extracts and tyrosinase inhibitors and free radical scavengers identified by HPLC post-column bioactivity assays, *Heliyon*, 9 (8) (2023) e18569–e18583.
- [24] J. Yan, G. Zhang, J. Pan, Y. Wang, α -Glucosidase inhibition by luteolin: kinetics, interaction and molecular docking, *Int. J. Biol. Macromol.* 64 (2014) 213–223.
- [25] C. Brusq, R. Miesusset, S.M. Hamdi, Development of a multivariable prediction model for congenital unilateral absence of the vas deferens in male partners of infertile couples, *Andrology* 10 (2) (2022) 262–269.
- [26] M.G. Fronza, D. Alves, D. Praticò, L. Savegnago, The neurobiology and therapeutic potential of multi-targeting β -secretase, glycogen synthase kinase β and acetylcholinesterase in Alzheimer's disease, *Ageing Res. Rev.* 90 (2023) 102033–102052.
- [27] J. Chen, M. Ran, M. Wang, X. Liu, S. Liu, Z. Ruan, N. Jin, Evaluation of antityrosinase activity and mechanism, antioxidation, and UV filter properties of theaflavin, *Biotechnol. Appl. Biochem.* 69 (3) (2022) 951–962.
- [28] Y.S. Zhang, J.S. Gong, Z.Y. Yao, J.Y. Jiang, C. Su, H. Li, C.L. Kang, L. Liu, Z.H. Xu, J. S. Shi, Insights into the source, mechanism and biotechnological applications of hyaluronidases, *Biotechnol. Adv.* 60 (2022) 108018–108035.
- [29] J. George, R. Stern, Serum hyaluronan and hyaluronidase: very early markers of toxic liver injury, *Clin. Chim. Acta* 348 (1–2) (2004) 189–197.
- [30] Z.-Z. Hu, T.-X. Ma, X.-M. Sha, L. Zhang, Z.-C. Tu, Improving tyrosinase inhibitory activity of grass carp fish scale gelatin hydrolysate by gastrointestinal digestion: Purification, identification and action mechanism, *Lwt* 159 (2022) 113205–113215.
- [31] Q. Wang, R. Li, N. Li, Y. Jia, Y. Wang, Y. Chen, P. Panichayupakaranant, H. Chen, The antioxidant activities, inhibitory effects, kinetics, and mechanisms of artocarpin and alpha-mangostin on α -glucosidase and α -amylase, *Int. J. Biol. Macromol.* 213 (2022) 880–891.
- [32] O. Bahadır Acikara, M. İlhan, E. Kurtul, K. Smejkal, E. Kupeli Akkol, Inhibitory activity of *Podospermum canum* and its active components on collagenase, elastase and hyaluronidase enzymes, *Bioorg. Chem.* 93 (2019) 103330–103335.
- [33] C. Mit, C. Tebby, T. Gueanno, A. Bado-Nilles, R. Beaudouin, Modeling acetylcholine esterase inhibition resulting from exposure to a mixture of atrazine and chlorpyrifos using a physiologically-based kinetic model in fish, *Sci. Total Environ.* 773 (2021) 144734–144743.
- [34] Y. Chen, M. Li, J. Lu, B. Chen, Influence of residential indoor environment on quality of life in China, *Build. Environ.* 232 (2023) 110068–110077.
- [35] A. Cincinelli, T. Martellini, Indoor Air Quality and Health, *Int. J. Environ. Res. Public Health* 14 (11) (2017) 1286–1290.
- [36] Y. Chen, B. Chen, J.F. Lu, Influence of residential indoor environment on self-rated health in China: a cross-sectional study, *Sci. Technol. Built Environ.* 28 (7) (2022) 916–927.
- [37] I.Y.S. Chan, A.M.M. Liu, Effects of neighborhood building density, height, greenspace, and cleanliness on indoor environment and health of building occupants, *Build. Environ.* 145 (2018) 213–222.
- [38] B. Su, R. Jadresin Milic, P. McPherson, L. Wu, Thermal Performance of School Buildings: Impacts beyond Thermal Comfort, *Int. J. Environ. Res. Public Health* 19 (10) (2022) 5811–5829.
- [39] C. Chen, J. An, C. Wang, X. Duan, S. Lu, H. Che, M. Qi, D. Yan, Deep Reinforcement Learning-Based Joint Optimization Control of Indoor Temperature and Relative Humidity in Office Buildings, *Buildings* 13 (2) (2023) 438–457.
- [40] G. Liu, M. Xiao, X. Zhang, C. Gal, X. Chen, L. Liu, S. Pan, J. Wu, L. Tang, D. Clements-Croome, A review of air filtration technologies for sustainable and healthy building ventilation, *Sustain. Cities Soc.* 32 (2017) 375–396.
- [41] M. Piasecki, K. Kostyrko, S. Pykacz, Indoor environmental quality assessment: Part I: Choice of the indoor environmental quality sub-component models, *J. Build. Phys.* 41 (3) (2017) 264–289.
- [42] Q. Deng, R. Wang, Y. Li, Y. Miao, J. Zhao, Human thermal sensation and comfort in a non-uniform environment with personalized heating, *Sci. Total Environ.* 578 (2017) 242–248.
- [43] W. Luo, R. Kramer, Y. de Kort, P. Rense, W. van Marken Lichtenbelt, The effects of a novel personal comfort system on thermal comfort, physiology and perceived indoor environmental quality, and its health implications - Stimulating human thermoregulation without compromising thermal comfort, *Indoor Air* 32 (1) (2022) e12951–e12967.
- [44] H. Sun, P. Saedi, S. Karuranga, M. Pinkepank, K. Ogurtsova, B.B. Duncan, C. Stein, A. Basit, J.C.N. Chan, J.C. Mbanya, M.E. Pavkov, A. Ramachandran, S.H. Wild, S. James, W.H. Herman, P. Zhang, C. Bommer, S. Kuo, E.J. Boyko, D.J. Magliano, IDF Diabetes Atlas: Global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045, *Diabetes Res. Clin. Pract.* 183 (2022) 109119–109131.
- [45] A.S. Alqahtani, S. Hidayathulla, M.T. Rehman, A.A. ElGamal, S. Al-Massarani, V. Razmovski-Naumovski, M.S. Alqahtani, R.A. El Dib, M.F. AlAjmi, Alpha-Amylase and Alpha-Glucosidase Enzyme Inhibition and Antioxidant Potential of 3-Oxolupenol and Katonic Acid Isolated from *Nuxia oppositifolia*, *Biomolecules* 10 (1) (2019) 61–79.
- [46] R.S. Tavares, S. Escada-Rebello, M.I. Sousa, A. Silva, J. Ramalho-Santos, S. Amaral, Can Antidiabetic Drugs Improve Male Reproductive (Dys)Function Associated with Diabetes? *Curr. Med. Chem.* 26 (22) (2019) 4191–4222.

- [47] U. Ghani, Re-exploring promising α -glucosidase inhibitors for potential development into oral anti-diabetic drugs: Finding needle in the haystack, *Eur. J. Med. Chem.* 103 (2015) 133–162.
- [48] J. Dowarah, V.P. Singh, Anti-diabetic drugs recent approaches and advancements, *Bioorg. Med. Chem.* 28 (5) (2020) 115263–115340.
- [49] D. Goyal, S. Shuaib, S. Mann, B. Goyal, Rationally Designed Peptides and Peptidomimetics as Inhibitors of Amyloid- β (A β) Aggregation: Potential Therapeutics of Alzheimer's Disease, *ACS Comb. Sci.* 19 (2) (2017) 55–80.
- [50] G. Li, G. Hong, X. Li, Y. Zhang, Z. Xu, L. Mao, X. Feng, T. Liu, Synthesis and activity towards Alzheimer's disease in vitro: Tacrine, phenolic acid and ligustrazine hybrids, *Eur. J. Med. Chem.* 148 (2018) 238–254.
- [51] S. Lista, A. Vergallo, S.J. Teipel, P. Lemercier, F.S. Giorgi, A. Gabelle, F. Garaci, N. B. Mercuri, C. Babiloni, B.P. Gaire, Y. Koronyo, M. Koronyo-Hamaoui, H. Hampel, R. Nistico, I. Neurodegeneration Precision Medicine, Determinants of approved acetylcholinesterase inhibitor response outcomes in Alzheimer's disease: relevance for precision medicine in neurodegenerative diseases, *Ageing Res. Rev.* 84 (2023) 101819–101840.
- [52] L.J. Walczak-Nowicka, M. Herbet, Acetylcholinesterase Inhibitors in the Treatment of Neurodegenerative Diseases and the Role of Acetylcholinesterase in their Pathogenesis, *Int. J. Mol. Sci.* 22 (17) (2021) 9290–9352.
- [53] D.S.C. Sharma, M.A. Lahiri, Use of hyaluronidase in plastic surgery: A review, *J. Plast. Reconstr. Aesthet. Surg.* 74 (7) (2021) 1610–1614.
- [54] E. Papakonstantinou, M. Roth, G. Karakiulakis, Hyaluronic acid: A key molecule in skin aging, *Dermato-Endocrinology* 4 (3) (2014) 253–258.
- [55] M. Yamamoto, M. Fronza, G.F. Caetano, M.N. Leite, C.S. Bitencourt, F.W.G. Paula-Silva, T.A.M. Andrade, M.A.C. Frade, I. Merfort, L.H. Faccioli, Hyaluronidase Modulates Inflammatory Response and Accelerates the Cutaneous Wound Healing, *PLoS One* 9 (11) (2014) e112297–e112308.
- [56] S.H. Bailey, S. Fagien, R.J. Rohrich, Changing Role of Hyaluronidase in Plastic Surgery, *Plast. Reconstr. Surg.* 133 (2) (2014) 127e–132e.
- [57] X. Huang, L. Zhang, F. Zheng, Y. Fang, F. Gao, Q. Chen, Q. Zeng, W. Zeng, Debut of a novel AIE-based fluorescent probe as tyrosinase tracer to image skin pigmentary disorders, *Sens. Actuators B* 389 (2023) 133889–133900.
- [58] Q. Chen, C. Shang, M. Han, C. Chen, W. Tang, W. Liu, Inhibitory mechanism of scutellarein on tyrosinase by kinetics, spectroscopy and molecular simulation, *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 296 (2023) 122644–122654.
- [59] X.-Z. Wu, W.-J. Zhu, L. Lu, C.-M. Hu, Y.-Y. Zheng, X. Zhang, J. Lin, J.-Y. Wu, Z. Xiong, K. Zhang, X.-T. Xu, Synthesis and anti- α -glucosidase activity evaluation of betulinic acid derivatives, *Arab. J. Chem.* 16 (5) (2023) 104659–104672.
- [60] A. Pitchai, R.K. Rajaretnam, R. Mani, N. Nagarajan, Molecular interaction of human acetylcholinesterase with trans-tephrostachin and derivatives for Alzheimer's disease, *Heliyon*, 6 (9) (2020) e04930–e04941.
- [61] A.Y. Ragheb, M.A. El-Ansari, O.A. Heikal, A.F. Galal, A.A.A. Salama, M.E. S. Kassem, N.A.M. Saleh, Phenolic glycosides and bioactive *Mangifera indica* L. kernel extract as neuroprotective agents against LPS-induced Alzheimer's disease in rats, *S. Afr. J. Bot.* 157 (2023) 37–43.
- [62] M. Guo, M. Zhou, B. Li, C. Du, R. Yao, L. Wang, X. Yang, W. Yu, Reducing indoor relative humidity can improve the circulation and cardiorespiratory health of older people in a cold environment: A field trial conducted in Chongqing, China, *Sci. Total Environ.* 817 (2022) 152695–152702.
- [63] P. Wolkoff, Indoor air humidity, air quality, and health – An overview, *Int. J. Hyg. Environ. Health* 221 (3) (2018) 376–390.
- [64] Q. Cheng, L. Bai, Y. Zhang, H. Zhang, S. Wang, M. Xie, D. Zhao, H. Su, Ambient temperature, humidity and hand, foot, and mouth disease: A systematic review and meta-analysis, *Sci. Total Environ.* 625 (2018) 828–836.
- [65] X. Hou, Z. Mao, X. Song, N. Kang, C. Zhang, R. Li, Y. Yuchi, W. Liao, X. Liu, W. Huo, C. Wang, J. Hou, Kitchen ventilation alleviated adverse associations of domestic fuel use and long-duration cooking with platelet indices as biomarkers of cardiovascular diseases, *Sci. Total Environ.* 834 (2022) 155341–155349.
- [66] W. Liao, X. Liu, N. Kang, Y. Song, R. Li, X. Song, X. Hou, C. Zhang, W. Huo, Z. Mao, J. Hou, C. Wang, Effect modification of kitchen ventilation on the associations of solid fuel use and long-duration cooking with the increased prevalence of depressive and anxiety symptoms: The Henan Rural Cohort Study, *Indoor Air* 32 (3) (2022) e13016–e13025.
- [67] J. Razjouyan, H. Lee, B. Gilligan, C. Lindberg, H. Nguyen, K. Canada, A. Burton, A. Sharafkhan, K. Srinivasan, F. Currim, S. Ram, M.R. Mehl, N. Goebel, M. Lunden, S. Bhangar, J. Heerwagen, K. Kampschroer, E.M. Sternberg, B. Najafi, Wellbuilt for wellbeing: Controlling relative humidity in the workplace matters for our health, *Indoor Air* 30 (1) (2020) 167–179.
- [68] R. Martin, E.M.O.R. Group, Symptoms of dry eye related to the relative humidity of living places, *Cont. Lens Anterior Eye* (2023) 101865–101871.
- [69] M. Qin, P. Hou, Z. Wu, J. Wang, Precise humidity control materials for autonomous regulation of indoor moisture, *Build. Environ.* 169 (2020) 106581–106617.
- [70] K. Tsuzuki, K. Okamoto-Mizuno, K. Mizuno, Effects of humid heat exposure on sleep, thermoregulation, melatonin, and microclimate, *J. Therm. Biol.* 29 (1) (2004) 31–36.
- [71] H. Tsutsumi, S.-I. Tanabe, J. Harigaya, Y. Iguchi, G. Nakamura, Effect of humidity on human comfort and productivity after step changes from warm and humid environment, *Build. Environ.* 42 (12) (2007) 4034–4042.
- [72] W. Liao, X. Liu, N. Kang, Y. Song, L. Wang, Y. Yuchi, W. Huo, Z. Mao, J. Hou, C. Wang, Associations of cooking fuel types and daily cooking duration with sleep quality in rural adults: Effect modification of kitchen ventilation, *Sci. Total Environ.* 854 (2022) 158827–158835.